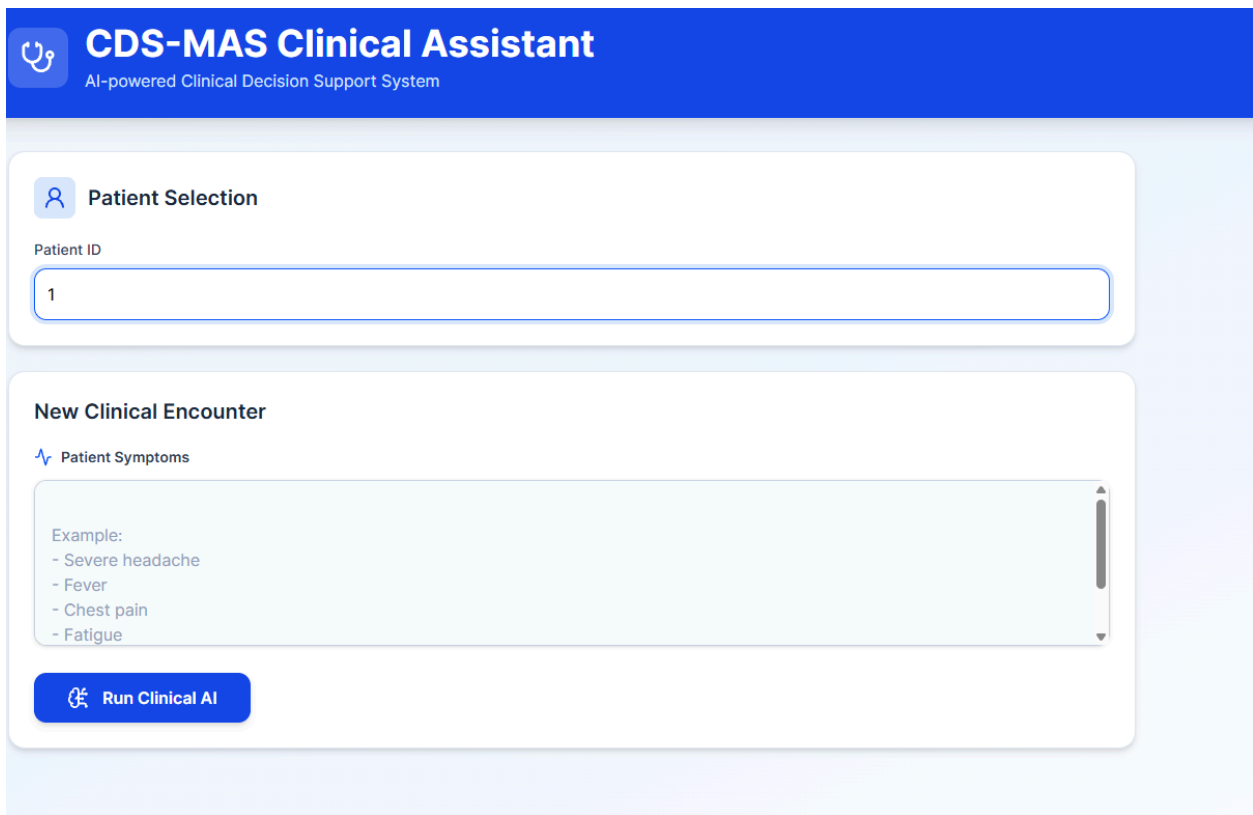
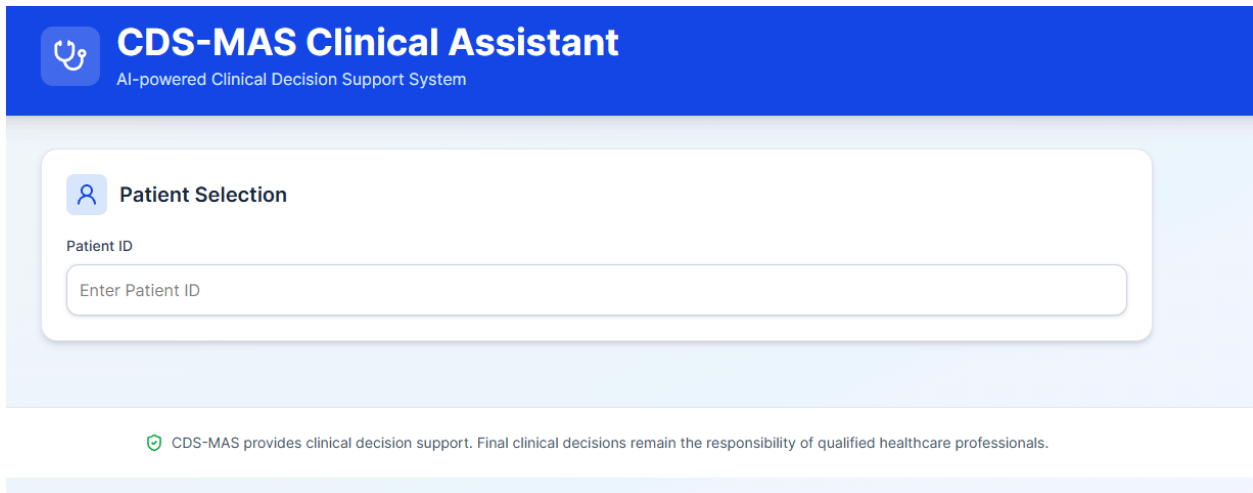


Screenshots

Full System in Action Screen Shorts





CDS-MAS Clinical Assistant

AI-powered Clinical Decision Support System



Patient Selection

Patient ID

1

New Clinical Encounter

Patient Symptoms

Severe headache



Running AI Analysis...

Clinical agents are analysing patient information...



Triage Assessment

1. Urgency Level: Emergency (Given the recurrent severe headaches and chest pain, immediate assessment is warranted.)
2. Clinical Concerns: Recurrent Severe Headache, Chest Pain, Dizziness, Persistent Symptoms Across Multiple Encounters without Documented Chief Complaints or Diagnoses
3. Recommended Actions: Immediate neurological and cardiovascular evaluation including but not limited to vital signs monitoring, ECG, possible imaging studies (CT/MRI of the head), blood tests for infection markers, complete blood count, electrolytes, coagulation profile, as well as a chest X-ray or CT angiography if indicated.
4. Missing Information Required: Chief Complaint(s) from previous encounters with documented chief complaints and diagnoses; detailed patient history including family medical history, personal risk factors (smoking status, alcohol use), medication list, allergies, recent travel or exposures to infectious diseases.

Clinical Knowledge Support

1. Clinical Considerations: The recurrent severe headaches and chest pain, especially when associated with dizziness as noted in the previous encounters of simplicio sithole, raise concerns for potential cardiovascular or neurological conditions that may require urgent attention. These symptoms could be indicative of a range of issues from migraines to more serious conditions such as myocardial infarction (heart attack) or cerebrovascular accidents (strokes).

2. Potential Risks: The combination of severe headache, chest pain, and dizziness can be symptoms associated with life-threatening emergencies such as myocardial infarction, stroke, or other cardiovascular events that require immediate intervention to prevent morbidity and mortality.

3. Recommended Evaluation: Given the urgency of these symptoms, a comprehensive evaluation is recommended which should include vital signs monitoring (blood pressure, heart rate, respiratory rate, temperature), an electrocardiogram (ECG) to assess for any cardiac abnormalities, blood tests including complete blood count, coagulation profile, and markers of infection or inflammation. Imaging studies such as a CT scan of the head may be necessary if there is suspicion of neurological causes like hemorrhage or stroke.

4. Knowledge Limitations: The information provided lacks documented chief complaints from previous encounters, detailed patient history including family medical history and personal risk factors such as smoking status, alcohol use, medication list, allergies, recent travels, exposures to infectious diseases or any prior diagnostic test results. This missing information is crucial for a comprehensive assessment of the current symptoms in relation to past medical history and potential risk factors that could contribute to his presenting condition.

Generated Clinical Note

PATIENT INFORMATION

=====

Name: Simplicio Sithole

Gender: Male (Assumed based on the name)

Phone: +263787456767

Not documented. Age not provided, assumed to be necessary for a complete clinical assessment but is unavailable from given information.

CHIEF COMPLAINT

=====

Severe headache (Current Symptom) and chest pain with dizziness across multiple encounters without definitive diagnoses or documented chief complaints at the time of each encounter, as per previous triage output data provided by AI.

HISTORY OF PRESENT ILLNESS

=====

Not documented in detail within this context; however, it is known that patient has experienced severe headache and chest pain multiple times over a short period with associated dizziness as noted from the triage output data provided by AI. No definitive diagnoses have been recorded for these symptoms across previous encounters based on available information.

RELEVANT HISTORY

=====

Not documented in detail within this context; however, it is known that patient has experienced severe headache and chest pain multiple times over a



Generated Clinical Note

RELEVANT HISTORY

=====

Not documented in detail within this context; however, it is known that patient has experienced severe headache and chest pain multiple times over a short period with associated dizziness as noted from the triage output data provided by AI. No definitive diagnoses have been recorded for these symptoms across previous encounters based on available information.

CLINICAL ASSESSMENT

=====

Not documented in detail within this context; however, it is known that patient has experienced severe headache and chest pain multiple times over a short period with associated dizziness as noted from the triage output data provided by AI. No definitive diagnoses have been recorded for these symptoms across previous encounters based on available information.

RECOMMENDED NEXT STEPS

=====

Immediate neurological and cardiovascular evaluation is recommended, including but not limited to vital signs monitoring, ECG, possible imaging studies (CT/MRI of the head), blood tests for infection markers, complete blood count, electrolytes, coagulation profile. A chest X-ray or CT angiography may be indicated based on further assessment findings and clinical judgment as per triage output data provided by AI.

NOTE: The patient has not been physically examined; therefore, the above recommendations are for supportive decision making only until a physical evaluation can take place under appropriate medical supervision to confirm any diagnoses or rule out conditions that require immediate intervention.



Verified Clinical Data

=====

VERIFIED CLINICAL DATA

=====

Patient Information:

Name:

simplicio sithole

Gender:

Male

Unit Testing

```
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Simplicio Sithole\Desktop\medicalAI
cachedir: .pytest_cache
rootdir: C:\Users\Simplicio Sithole\Desktop\medicalAI
plugins: anyio-4.14.1, langsmith-0.9.8, asyncio-1.4.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=None
collected 4 items

tests/test_clinical_api.py::test_clinical_workflow PASSED
tests/test_database.py::test_create_encounter PASSED
tests/test_database.py::test_patient_history PASSED
tests/test_workflow.py::test_ai_workflow_execution PASSED
```

test_report.html

Report generated on 13-Jul-2026 at 18:48:28 by [pytest-html](#) v4.2.0

Environment

Python	3.12.10
Platform	Windows-11-10.0.22631-SP0
Packages	• pytest: 9.1.1 • pluggy: 1.6.0
Plugins	• anyio: 4.14.1 • langsmith: 0.9.8 • asyncio: 1.4.0 • html: 4.2.0 • metadata: 3.1.1

Summary

4 tests took 00:06:55.

(Un)check the boxes to filter the results.

☒ 0 Failed, ☒ 4 Passed, ☒ 0 Skipped, ☐ 0 Expected failures, ☒ 0 Unexpected passes, ☒ 0 Errors, ☐ 0 Reruns ☐ 0 Retried,

[Show all details](#) / [Hide all details](#)

Result 	Test	Duration	Links
Passed	tests/test_clinical_api.py::test_clinical_workflow	00:02:50	
Passed	tests/test_database.py::test_create_encounter	7 ms	
Passed	tests/test_database.py::test_patient_history	3 ms	
Passed	tests/test_workflow.py::test_ai_workflow_execution	00:03:36	

API Swagger Screenshot

default	
GET	/health Health Check
GET	/ Root
Clinical Workflow	
POST	/clinical/workflow Run Clinical Workflow
Patients	
GET	/patients/ Read Patients
POST	/patients/ Create New Patient
GET	/patients/{patient_id} Read Patient
PUT	/patients/{patient_id} Edit Patient

API Clinical Workflow Backend API point Testing

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/clinical/workflow' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "patient_id": 1,
    "symptoms": "Chest pain with shortness of breath",
    "patient_information": "45 year old male patient"
  }'
```

Request URL

http://127.0.0.1:8000/clinical/workflow

Server response

Code

Details

200

Response body

```
{
  "triage_result": "\nUrgency Level: Emergency\n\nPossible Concerns: The combination of chest pain and shortness of breath can be indicative of serious conditions such as myocardial infarction (heart attack), pulmonary embolism, or pneumothorax. These symptoms require immediate medical attention to prevent severe outcomes including heart damage or death if treatment is delayed.\n\nRecommended Next Steps: Advise the patient that they need to go directly to an emergency department right away for a thorough evaluation and monitoring of cardiovascular function, as this situation potentially could be life-threatening without prompt medical intervention.\n\nDo not provide diagnostic confirmation or detailed prognosis in triage communication; instead focus on the urgency and immediate need for professional assessment at a healthcare facility equipped to handle potential emergencies such as cardiology centers, ERs with specialized care units (e.g., catheterization labs), or facilities that can perform necessary imaging studies like CT pulmonary angiograms on the spot if time permits and resources are available. \n\nRemember to maintain a calm and reassuring tone while communicating urgency, emphasizing prompt action without causing panic: \"Your symptoms require immediate evaluation by medical professionals who can provide you with all the care that's needed right now.\"",
  "knowledge_result": "\nRelevant Clinical Considerations for the presented case include possible life-threatening conditions like acute coronary syndrome (heart attack), pulmonary embolism, or pneumothorax due to tracheal deviation and hemoptysis symptoms. The patient's history of presenting chest pain with associated sweating may also suggest a panic attack but should not be ruled out without proper assessment given the serious nature of other potential conditions mentioned above.\n\nPossible Risks: If immediate medical attention is delayed, there are significant risks including myocardial infarction leading to irreversible heart damage or even death if blood supply compromise continues unaddressed; pulmonary embolism may result in fatal respiratory distress and right heart strain due to blockage of the pulmonary artery, while a pneumothorax could lead to progressive collapse requiring chest tube insertion.\n\nRecommended Next Steps: Given these symptoms' potential severity, it is crucial that you seek immediate medical care at an emergency department equipped with cardiology and pulmonary facilities (catheterization labs). Inquire about the availability of a CT pulmonary angiogram if time permits. It's important to remain calm yet assertive, emphasizing your symptoms warrant urgent care without causing unnecessary alarm: \"I understand these feelings are distressing, but it is essential that I receive immediate medical attention for thorough evaluation and timely treatment.\"",
  "clinical_note": "\n*1. Chief Complaint:* The chief complaint of the patient, who reports that he has been experiencing chest pain accompanied by sweating for an unknown duration, which started while performing physical labor without any known injury or trigger event. He also notes a recent onset of hemoptysis (coughing up blood).\n\n*2. History of Present Illness:* The patient's history includes no past medical problems except mild seasonal allergies, which he manages with over-the-counter antihistamines as needed. He denies any family history of heart disease or clotting disorders but admits to a sedentary lifestyle and being currently under significant work-related stress without routine physical activity for several months due to worsening company financial status impacting his job security, increasing anxiety about potential unemployment.\n\n*3. Relevant Findings:* Physical examination reveals tracheal deviation towards the right side and decreased breath sounds on that same lung field; however, there are no signs of peripheral cyanosis or jugular venous distension to suggest superior vena cava syndrome (SVC). The patient appears anxious but oriented.\n\n*4. Assessment:* Based upon the symptoms and clinical findings presented - chest pain, hemoptysis with no prior trauma history, radiating discomfort down both arms followed by diaphoresis during physical exertion (despite this being his first episode), along with a deviated trachea on examination; there is an immediate concern for life-threatening cardiovascular or pulmonary conditions. These could include acute coronary syndrome, possibly myocardial infarction, due to reported chest pain and associated sweating (possibly sympathetic activation); a potential massive embolism that has caused either an occlusion leading to secondary infarction or a pulmonary hypertension scenario; though this would be less likely without any past history. A pneumothorax should also remain on the differential diagnosis, but tracheal deviation and hemoptysis are more consistent with other conditions if not iatrogenic due to..."
```

Download

Response headers

```
content-length: 6172
content-type: application/json
date: Fri, 10 Jul 2026 10:39:20 GMT
server: uvicorn
```